

GNN Testing Report

Propagation-confidence GNN, NSE Auto supply-chain graph · generated 15 Sep 2026 · suite: 76 passed, 0 failed, 1 skipped

1. What is being tested

The GNN is one of the three models in the trading system. It looks at the 15-company supply-chain graph (32 curated supplier-to-buyer links) and, for each directed pair of companies $A \rightarrow B$, estimates the probability that a *shock* at A today will be followed by a shock at B within the next five trading days. A shock is an unusually large one-day price move. These probabilities are computed once for every trading day, cached to disk, and fed into the reinforcement-learning agent's state. The RL agent never calls the GNN directly.

Testing therefore has to answer four questions: (1) are the training labels built correctly, (2) is the data reaching the model in the right shape with no leakage from the future, (3) does the model architecture behave the way a graph model should, and (4) does the trained checkpoint actually contain signal that a trivial rule of thumb would not give. Section 3 explains each test in plain words; Section 2 analyses what the trained model does.

2. Results analysis

2.1 Headline numbers

Metric	Value
Validation AUC (trained checkpoint, recomputed)	0.6002
Validation AUC recorded when training finished	0.6002
Training AUC	0.6721
Train – val gap	0.0719
Baseline: train-period return correlation of the pair	0.5269
Baseline: buyer's previous-day volatility (raw AUC)	0.4868
Rule: buyer's previous-day volatility, best direction	0.5132
Baseline: supplier's previous-day volatility (raw AUC)	0.4772
Validation positive rate (a coin-flip scores 0.5 AUC)	16.8%
Early-stopping epoch	3

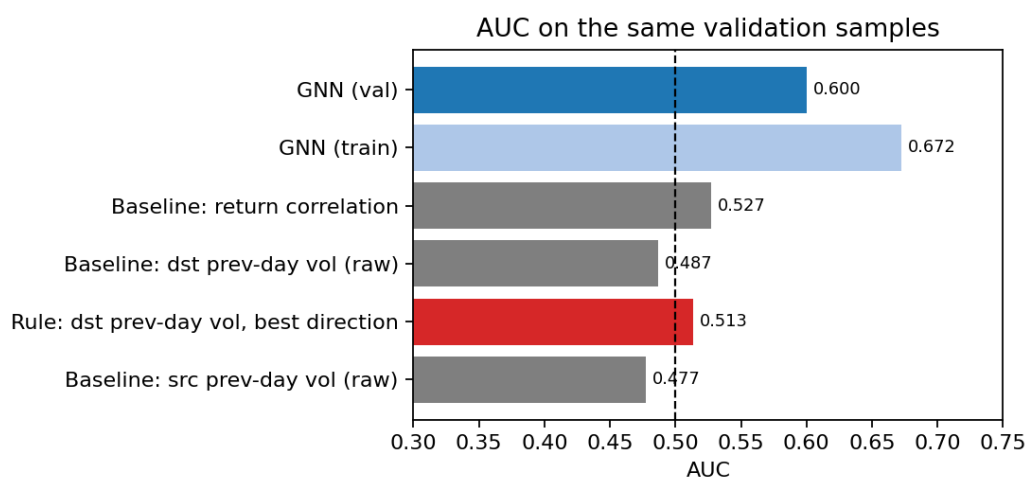


Figure 1 — AUC of the GNN and the no-learning baselines on the identical 1206 validation samples. A ranking rule can be used either way round, so the red bar is the volatility rule in whichever direction works.

AUC (area under the ROC curve) measures ranking quality: 1.0 means every real cascade is scored above every non-cascade, 0.5 means random. The GNN scores 0.600. The strongest baseline — simply looking up how correlated the two stocks' returns were during the training period — scores 0.527. So the GNN adds +0.073 AUC on top of "pairs that co-move tend to cascade". The volatility rule — rank pairs by the buyer's previous-day

volatility, low first — scores 0.513, which the GNN clears by +0.087. This rule matters because under the project's original label (raw returns against a 21-day sigma) it scored 0.64 out-of-sample and beat every model tested — a calm stock cleared 2.5× its stale sigma as soon as normal volatility returned, so the "cascades" were largely a volatility-regime artifact. The label was therefore redefined on sector-excess returns (stock minus Nifty Auto) against a 63-day sigma, under which the rule scores about 0.52 out-of-sample (see section 2.8).

2.2 Data under test

Item	Value
Timeline	3635 trading days, 2011-10-03 -> 2026-06-30
Graph	15 companies, 32 curated edges, 64 scored directed pairs (both directions)
Node features	logret_1, logret_5, logret_21, vol_21, mom_63, volz_21, present (7)
Pair-head inputs	rolling_corr, edge_weight, direction
Shock label	sector-excess 1-day return > max(2.5 × previous-day 63d sigma, 2%); cascade within 5 days
Training samples	4357 (16.9% positive), 2012-01-06 -> 2023-04-10
Validation samples	1206 (16.8% positive), 2023-05-12 -> 2025-12-03
Shock-day rate	2.70% of all (day, company) cells
Model	graphconv, hidden 64, 2 layers, dropout 0.4, GRU over 10 days, seed 42

Validation is the chronological tail of the timeline with a five-day gap before it, so no training label window overlaps the validation period. The per-company shock counts run from 34 to 110 shock days over the timeline (SONACOMS only lists in 2021), so the volatility-scaled threshold gives every stock a comparable number of events.

2.3 Is the signal stable over time?

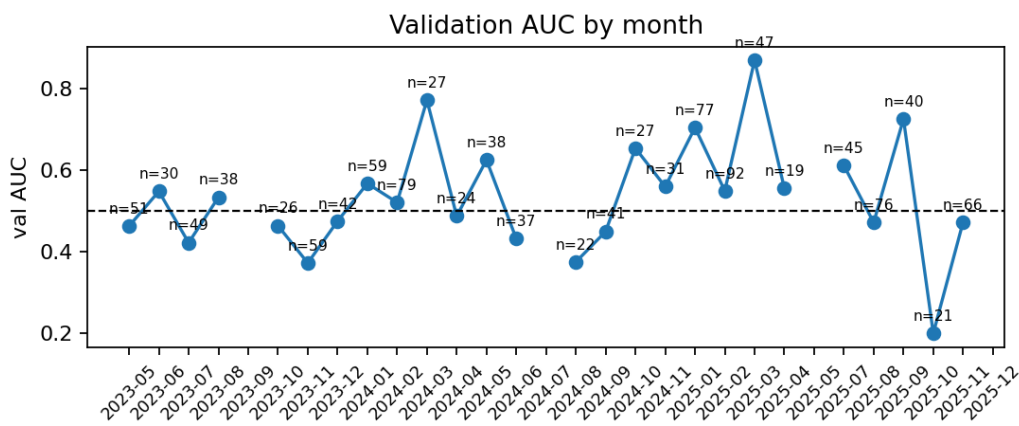


Figure 2 — validation AUC computed month by month (n = samples in that month).

No. The month-by-month AUC ranges from 0.20 (2025-10) to 0.87 (2025-03), and 12 of 26 months fall below 0.5 (2023-05, 2023-07, 2023-10, 2023-11, 2023-12, 2024-04, 2024-06, 2024-08, 2024-09, 2025-08, 2025-10, 2025-11). Monthly sample sizes are small (19–92) so a lot of this is noise, but the practical reading is that the GNN's score is an intermittent signal, not a steady one. The RL agent has to learn how much to trust it, which is exactly what the 'no-gnn' ablation arm will measure.

2.4 Direction, calibration and features

Direction	n	positive rate	AUC
Downstream (supplier → buyer)	619	14.9%	0.581
Upstream (buyer → supplier)	587	18.9%	0.618

Both directions score about equally, which justifies the decision to supervise both (supply disruptions travel downstream to the car maker; demand shocks travel upstream to the supplier). Neither half is dead weight.

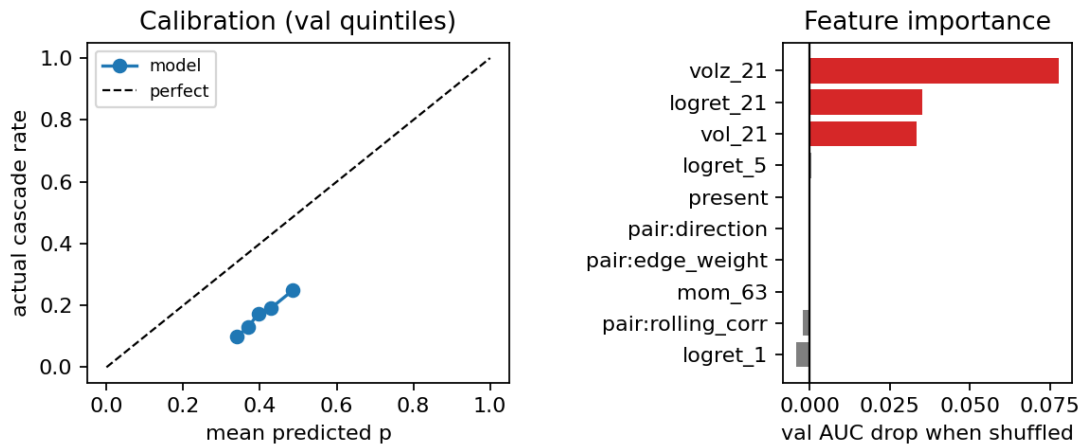


Figure 3 (left) — calibration: mean predicted probability per quintile vs actual cascade rate. Figure 4 (right) — permutation importance: validation AUC drop when one input feature is shuffled across days.

Calibration. The five quintiles are ordered correctly: the lowest-scored fifth of pairs cascaded 10% of the time, the highest-scored fifth 25%. The absolute probabilities are inflated (mean predicted 0.40 against a 17% base rate) because training re-weights the rare positive class; this is intended. The score is a ranking, not a calibrated probability, and the RL state treats it as such.

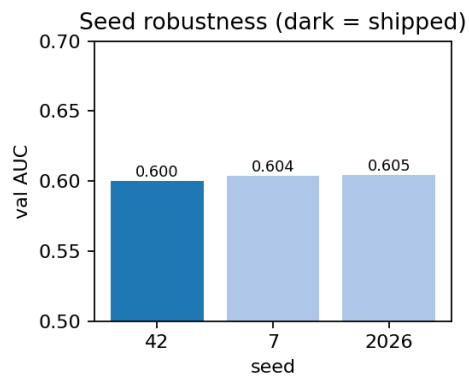
Features. Shuffling **volz_21** costs 0.078 AUC; **logret_21**, **vol_21** also matter. Inputs whose shuffling costs under 0.01 AUC: logret_5, present, pair:edge_weight, pair:direction, mom_63, pair:rolling_corr, logret_1. Shuffling one input at a time understates inputs that carry the same information as another, so this is a guide to what the model leans on, not a verdict on what is useless.

2.5 Per-pair view

Pair	n	pos	AUC
BAJAJ-AUTO.NS->EXIDEIND.NS	13	1	1.00
SONACOMS.NS->TMPV.NS	18	1	1.00
UNOMINDA.NS->MARUTI.NS	12	1	1.00
EXIDEIND.NS->TMPV.NS	21	1	0.95
...			
BHARATFORG.NS->ASHOKLEY.NS	19	4	0.38
MARUTI.NS->UNOMINDA.NS	16	1	0.33
TMPV.NS->BOSCHLTD.NS	15	3	0.33
MOTHERSON.NS->TMPV.NS	17	1	0.19

61 pairs have enough validation samples to score; 14 are below 0.5. The weak pairs cluster around **BOSCHLTD.NS** (5 of them). With roughly nine samples per pair this is a hint rather than a verdict, but it points at that node's curated edges in relationships.csv as the first thing to re-examine.

2.6 Seed robustness



Retraining from scratch with 2 other random seeds gives 0.600–0.605. The shipped seed (42) ranks 3 of 3 and stopped at epoch 3. The spread across seeds (0.005) is smaller than the +0.073 margin over the correlation baseline; quote the headline AUC as "about 0.60–0.60", not to three decimals. Retrains went to a temporary folder; the shipped weights were not touched.

2.7 Reproducibility and leakage

The cached score file used by the RL agent (3635 days × 64 pairs) was recomputed from the weights: maximum difference 0.0e+00, i.e. bit-for-bit identical. Corrupting every day *after* day t with noise leaves day t's scores exactly unchanged, while nudging a day *inside* the ten-day window does change them. Together these prove the scores the agent sees are deterministic and contain no information from the future. The train/val gap of 0.072 with early stopping at epoch 3 shows no overfitting.

2.8 Out-of-sample: walk-forward comparison

The validation AUC above is one split. The stronger test is walk-forward: five half-year windows (2023H2–2025H2), each scored by a model retrained on everything before it, three seeds averaged, 1131 samples over 183 shock days; 2026 (the PPO test window) is never used. Confidence intervals resample whole shock days, because pairs scored on the same day are not independent. "Vol-strat" is the mean AUC inside quintiles of the buyer's previous-day volatility: what a score adds once that volatility is held fixed. The shipped configuration is variant V4.

Variant	Description	AUC	95% CI	Vol-strat	P(>V0)	P(>vol rule)
V0	previous model shape (2021+ data, SAGE, no pair features)	0.538	0.474–0.593	0.542	—	57%
V1	long history 2011+ (adjusted, masked)	0.582	0.531–0.639	0.579	83%	90%
V2	V1 + edge weights (GraphConv)	0.569	0.521–0.623	0.566	77%	90%
V3	V1 + pair features	0.589	0.530–0.646	0.582	88%	90%
V4 (shipped)	V1 + edge weights + pair features (idea 2)	0.584	0.535–0.636	0.574	86%	91%
V5	V1 + India context + breadth	0.564	0.482–0.633	0.560	76%	70%
V6	V1 + full context + breadth, PCA-8	0.515	0.456–0.580	0.507	35%	37%
V7	V4 + India context + breadth	0.566	0.494–0.632	0.564	79%	74%
V8	V4 + full context + breadth, PCA-8	0.565	0.510–0.625	0.556	74%	83%
V9	V4 + full context + breadth (raw 151)	0.500	0.430–0.566	0.494	11%	31%
neg_dst_vol	rule: lower previous-day 21d volatility of dst	0.525	0.463–0.597	0.500	43%	—
corr_2y	rule: trailing 2y return correlation of the pair	0.541	0.486–0.607	0.544	55%	—

Full tables: [reports/gnn_walkforward/WALKFORWARD.md](#). The earlier run under the raw-return label is archived beside it.

3. The tests, in plain words

The suite lives in `tests/gnn/` and runs with `pytest tests/gnn`. Each test is a small Python function that sets up a situation where the correct answer is known in advance, runs the real project code on it, and asserts the answer matches. If any assertion fails, `pytest` reports it. There are 77 tests in six groups. Tests in groups A–D use tiny hand-made inputs and need nothing on disk; groups E and F use the real data and the real trained weights and skip themselves if those are missing.

A. Shock detection and label construction (18 tests)

What is checked. A "shock" is a one-day move larger than $2.5\times$ the stock's recent (63-day) volatility, with a floor of 2% so that calm periods don't flag trivial moves, and the move is measured net of the Nifty Auto index (sector-excess return) so a market-wide day does not count as a shock at every company at once. A training example is "company A shocked today; did company B shock within the next 5 trading days?" These tests confirm that definition is implemented exactly, for both the raw-return and the sector-excess variants of the rule, and that the sigma window can differ from the feature window.

How. Each test builds a miniature price table for two fake companies (a few days, numbers chosen by hand) and calls the same shock and label functions the training code uses. For example: with 1% volatility yesterday, a 3% move must count as a shock and a 2% move must not; a 50% move on the very first day must *not* count, because there is no previous-day volatility to compare against; the volatility used must be yesterday's, not today's, so a big move cannot raise its own threshold. For labels: a shock at B on the same day as A is not a cascade; a shock exactly five days later is; six days later is not; shocks in the last five days of the timeline are dropped because their five-day window would be cut off. The train/validation split is checked to be strictly chronological with a five-day gap, and never shuffled.

B. Data-to-tensor bridge (13 tests)

What is checked. The feature table has to be turned into a 3-D array of [day, company, feature] in a fixed company order, and the model is fed a trailing ten-day window ending at the day being scored. Any mistake here silently corrupts everything downstream.

How. Small tables with recognisable numbers (1, 2, 3 for company A; 10, 20, 30 for company B) are converted and specific cells are checked. The windowing function is asked for a ten-day window at day 1 and must return two real days preceded by zero-padding, never a day after the target. On the real dataset the tests check the array shape ($3635 \times 15 \times 7$), that every curated edge has its reverse present for message passing, that the 64 scored pairs are exactly the 32 edges plus their reverses with no duplicates, that features were normalised using only training-period statistics (mean ≈ 0 , std ≈ 1 on training rows), and that the training/validation counts match the numbers recorded at training time.

C. Model architecture (15 tests)

What is checked. Properties any correct graph neural network must have, independent of data. These catch wiring bugs that would not necessarily crash but would make the model meaningless.

How. The model is built on random inputs and probed. Two models built with the same seed must give identical outputs (reproducibility). In evaluation mode two calls must agree exactly; in training mode they must differ (dropout is really on/off). After one backward pass every parameter must have a non-zero gradient (nothing is disconnected). Scoring four pairs together must equal scoring them one at a time (no cross-talk). $A \rightarrow B$ and $B \rightarrow A$ must get different scores (direction is respected). Renumbering the companies consistently in the features, the edges and the pairs must not change any score — this is the defining property of a graph network, called permutation equivariance. Reversing the order of the ten days must change the output (the recurrent layer is really order-aware), and deleting all edges must change it (the graph is really used). The alternative attention architecture builds and runs; an unknown architecture name is rejected; the parameter count equals the number the configuration implies.

D. Training utilities and a learning smoke test (7 tests)

What is checked. The AUC metric, the per-day batching, the class re-weighting, and — most importantly — that the full training loop can actually learn.

How. The AUC implementation is fed cases with known answers: perfectly ordered scores must give 1.0, perfectly reversed 0.0, all-equal 0.5, a single class must give "undefined", and on 500 random points it must agree with scikit-learn's reference to six decimals (skipped if scikit-learn is not installed). The batching helper is checked to group samples by day in date order. The class weight is checked to make one positive and one negative example carry equal total loss mass. The smoke test then builds a synthetic 8-company ring graph, plants a simple rule ("cascade if the supplier's first feature is positive"), and trains the real model for 60 steps: the loss must at least halve and the AUC must exceed 0.9. If gradients, batching or shapes were broken, this would fail.

E. The trained checkpoint (12 tests, real data)

What is checked. That the weights on disk are the ones described by their metadata, that they reproduce the recorded score, that they beat naive baselines, and that the cached scores consumed by the RL agent are exact and leak-free.

How. The metadata file is compared field by field to the configuration. Every tensor in the saved weights is checked for shape and for finite values. The saved normalisation statistics are compared to those recomputed from the data. The validation AUC is recomputed and must match the recorded 0.600 within 0.005. The correlation baseline and the volatility rule (taken in whichever direction scores higher) are computed on exactly the same validation samples and the GNN must beat both. The train/val gap must be under 0.2 and the scores must have spread (a constant output would pass everything else). The cache file must have the right columns, length, date index, no missing values and all values in $[0, 1]$, and five sampled days are recomputed from the weights and must match to five decimals. Finally the no-look-ahead test: score day 300, overwrite all later days with noise, score again — identical; nudge one day inside the window — different.

F. Extended inputs and corrected data (14 tests)

What is checked. Everything added when the training history was extended back to 2011: the demerger back-adjustment, the same-day / one-day-lag rules for outside market data, the presence mask for a company that did not exist yet, the pair-head inputs, the edge-weighted graph layer, and — above all — that none of it can see the future.

How. A five-day price table with a planted 40% demerger drop is adjusted and the ex-date return must equal only that day's real trading, with earlier returns unchanged. The as-of joiner is fed dated values and must return the same day for lag 0, the previous day for lag 1, and "missing" once a value is over a week stale. The GraphConv layer must change its output when edge weights change while GraphSAGE must not. On the real data: SONACOMS must be all-zero with presence 0 before its 2021 listing and never generate a sample before it; every tensor must be finite; the compression of outside market data must be fitted on training days only (scrambling every later day must not change it); the inputs built for live inference must equal the training inputs exactly; and for node features, market context and the rolling pair correlation alike, truncating all source data at day t must leave day t 's values unchanged. Finally, no training or validation label may read a price after the configured label end date, so the PPO test year never influences early stopping.

G. Diagnostics (not pass/fail)

The report script adds analyses that have no single right answer and are therefore reported rather than asserted: the baselines above; AUC split by direction and by month; calibration by quintile; permutation importance (shuffle one feature's values across days, re-score, measure the AUC drop); per-pair AUC; and seed robustness (retrain from scratch with different random seeds into a temporary folder and compare).

4. Conclusion

- **The code is correct.** All 76 tests pass. Labels, data plumbing, model mechanics and the training loop behave exactly as specified, and the checkpoint on disk reproduces its recorded metrics.
- **The pipeline is leak-free and reproducible.** Scores contain no future information and the cache is bit-identical to a fresh forward pass. This is the property that matters most for the downstream RL claim, and it is proven, not assumed.
- **The signal is real but modest.** Validation AUC 0.600 against 0.527 for a static correlation lookup, 0.513 for the volatility rule and 0.5 for chance. The input it leans on most is **volz_21**. Quoted honestly: the GNN adds a few points of ranking power over the simple rules; out-of-sample (walk-forward) the shipped configuration scores 0.584 against 0.525 for the volatility rule.
- **The signal is intermittent.** Monthly AUC swings between roughly 0.2 and 0.9. The RL agent should be expected to learn a modest, not dominant, reliance on it; the 'no-gnn' ablation will quantify that.
- **The headline number is seed-sensitive at the ± 0.01 level.** Seeds span 0.600–0.605. The shipped seed 42 is kept for reproducibility, per the project's fixed-seed rule; report the AUC as a range.

- **What changed and why.** The label moved from raw returns / 21-day sigma to sector-excess returns / 63-day sigma after the volatility rule was found to beat every model under the old definition; the training history was extended to 2011 with demerger-adjusted prices; the graph layer now uses the curated edge weights and the pair head sees the pair's rolling correlation. Each step was chosen on out-of-sample walk-forward AUC, never on the validation split reported above.
- **Next steps if time allows** (none block PPO training): re-examine the curated edges around the weakest node; re-run the seed sweep after any further feature change.

Sources: tests/gnn/ (suite), scripts/gnn_test_report.py (diagnostics, also writes GNN_TEST_REPORT.md), scripts/gnn_test_pdf.py (this document), notebooks/gnn_testing.ipynb (interactive version). pytest summary: 76 passed, 1 skipped in 6.97s.